

RL-Tuned Synthetic Marine Radar for Sim-to-Real Transfer

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CSCE 775: Deep Reinforcement Learning

Abstract—Machine-learning models for marine radar are often trained on synthetic data because real labeled logs are expensive to collect. A key challenge is the sim-to-real gap: models that perform well in simulation can fail on real sensor data. This proposal studies an environment-aware reinforcement learning controller that tunes simulator parameters using feedback from held-out real validation data, with environment labels such as lake, river, and coast. We compare the method against fixed randomization, Bayesian optimization, and unconditioned RL baselines, and evaluate both performance gains and sample efficiency under domain holdout and stress conditions.

I. INTRODUCTION

Marine radar is useful for tasks such as obstacle detection and local mapping, but training robust models requires large labeled datasets. Collecting and annotating real radar logs on water is expensive, so many pipelines rely heavily on synthetic radar generated in simulation. This creates a common sim-to-real gap: models that look strong in simulation can degrade when tested on real sensor data.

This proposal asks a simple question: can we automatically tune the simulator so synthetic training data better match the real world? We study a reinforcement learning (RL) controller that updates simulator parameters to maximize performance on held-out real radar validation data. The controller is conditioned on broad environment type (‘lake’, ‘river’, ‘coast’), with the hypothesis that this context-aware tuning will outperform fixed settings and uniform randomization, especially on unseen routes and weather/clutter conditions.

The method builds on domain randomization and adaptive sim-to-real transfer literature [1]–[13]. Prior work often uses fixed/random perturbations, one-shot adaptation, or calibration pipelines that do not explicitly model maritime environment type. Our contribution is an environment-aware RL outer loop that optimizes downstream real-task performance directly (Fig. 1), with explicit comparison against related methods (Fig. 2).

II. APPROACH

We cast simulator tuning as a bi-level optimization problem. Let $\mathcal{M}$ be the downstream perception model (e.g., MRISNet for ship segmentation [14]) and $\mathcal{S}(\theta, e)$ be a physically-grounded radar simulator parameterized by θ and conditioned on environment type $e \in \{\text{lake}, \text{river}, \text{coast}\}$. The objective is:

$$\theta^* = \arg \max_{\theta} \mathcal{R}(\mathcal{M}^*(\theta), \mathcal{D}_{real}) \quad (1)$$

subject to the inner-loop training:

$$\mathcal{M}^*(\theta) = \arg \min_{\omega} \mathcal{L}_{task}(\mathcal{M}_{\omega}, \mathcal{S}(\theta, e)) \quad (2)$$

where $\mathcal{L}_{task}$ is the segmentation loss on synthetic data and $\mathcal{R}$ is the validation reward on real data $\mathcal{D}_{real}$.

The outer-loop RL policy, $\pi(\theta_{t+1}|s_t)$, proposes sequential updates to θ . The state s_t includes the current parameter vector, the environment label e , and a history of validation performance. The reward function r_t is formulated as:

$$r_t = \Delta \text{mIoU}_{real} - \lambda_1 \|\theta_t - \theta_{init}\|_2 - \lambda_2 \mathbb{K}(\theta \notin \Omega_{phys}) \quad (3)$$

where $\Delta \text{mIoU}_{real}$ is the improvement in Mean Intersection over Union on the real validation set, and Ω_{phys} represents the physically plausible bounds of the simulator.

The simulator $\mathcal{S}$ employs ray-tracing to model multi-path interference and calculates the Radar Cross-Section (RCS) of maritime targets. To accurately represent dynamic sea clutter, we implement the Elfouhaily spectrum for sea surface modeling, accounting for non-Gaussian spiky clutter at low grazing angles. The policy controls physically meaningful factors including speckle strength, which governs the statistical distribution of coherent noise; the attenuation coefficient, for adjusting atmospheric and environmental signal loss; the clutter floor, which tunes baseline noise levels based on environment-specific wave heights; and multipath gain, for controlling the intensity of ghost reflections from shorelines.

We use Soft Actor-Critic (SAC) to optimize the policy, leveraging its maximum entropy framework for robust exploration of the high-dimensional parameter space. This environment-conditioned approach allows the policy to specialize tuning for different maritime contexts, significantly reducing the sim-to-real gap compared to uniform domain randomization.

III. EVALUATION

Evaluation uses real marine radar logs from the MOANA [15] and PoLaRIS [16] datasets, featuring explicit environment labels and strict splits by route/location to avoid leakage. We employ MRISNet [14] as the primary downstream perception model to evaluate ship instance segmentation performance. We compare five settings: (1) fixed simulator defaults, (2) uniform random domain randomization, (3) Bayesian optimization tuning, (4) RL tuning without environment labels, and (5) RL tuning with environment labels.

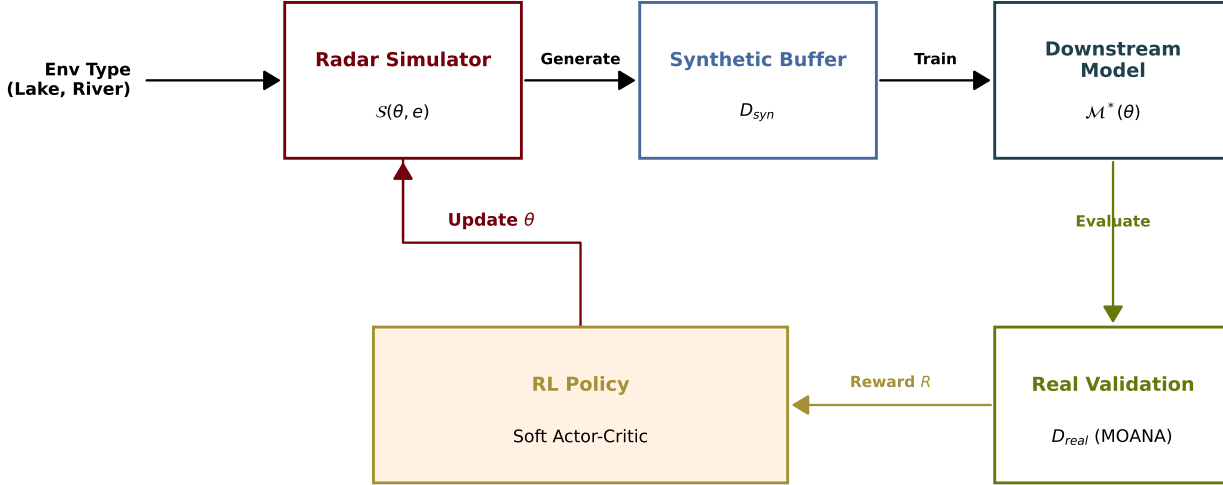


Fig. 1. Proposal 1 closed-loop method: an RL outer loop updates simulator parameters from real-validation reward, conditioned on environment type.

Method Family	Real-Feedback Adaptive	Env-Type Aware	Sequential Policy Control	Physical Bounds	Task-Reward Optimized
Fixed Domain Randomization (To-bin, Peng)	✗	✗	✗	✗	✗
SimOpt / BayesSim	✓	✗	✗	✗	✗
BayRn / ADR / CDR	✓	✗	✗	✗	✗
Radar Calibration (Digital Twin style)	✓	✗	✗	✓	✗
Proposed Method (Proposal 1)	✓	✓	✓	✓	✓

Fig. 2. Related-work capability comparison and the explicit improvement axes targeted by Proposal 1.

The primary outcome is real-data task performance and corresponding reduction in sim-to-real gap. Secondary outcomes are cross-domain generalization (train on two environment types, test on the held-out type), robustness on unseen locations, run-to-run variance, and behavior under low-SNR/high-clutter stress tests. We also report sample efficiency as the number of real-validation calls needed to reach a target accuracy.

Each novelty claim is tied to a direct test: (a) benefit of sequential RL adaptation versus BayRn under matched evaluation budgets [5]; (b) benefit of environment semantics via conditioned versus unconditioned RL; and (c) benefit of physical realism constraints via bound-violation and instability rates with/without realism regularization. This structure makes it explicit how the proposed method differs from existing randomization and adaptation families [1]–[4], [6], [7].

Prior published results already motivate this direction (Fig. 3). In real FMCW transfer, simulation-only baselines can approach chance-level behavior, while calibrated adaptation improves substantially [11]. In synthetic-to-measured SAR transfer, synthetic-only and simple-noise baselines are again weaker than stronger adaptation methods [10]. Together, these patterns motivate structured, feedback-driven simulator control rather than fixed randomization.

IV. CONCLUSION

This project targets a practical problem with broad relevance beyond radar: models trained in simulation often fail when deployed in the real world. We propose an environment-aware RL controller that tunes synthetic data generation to improve real-data task performance, and we benchmark it against strong non-RL alternatives under realistic transfer stress tests. If full

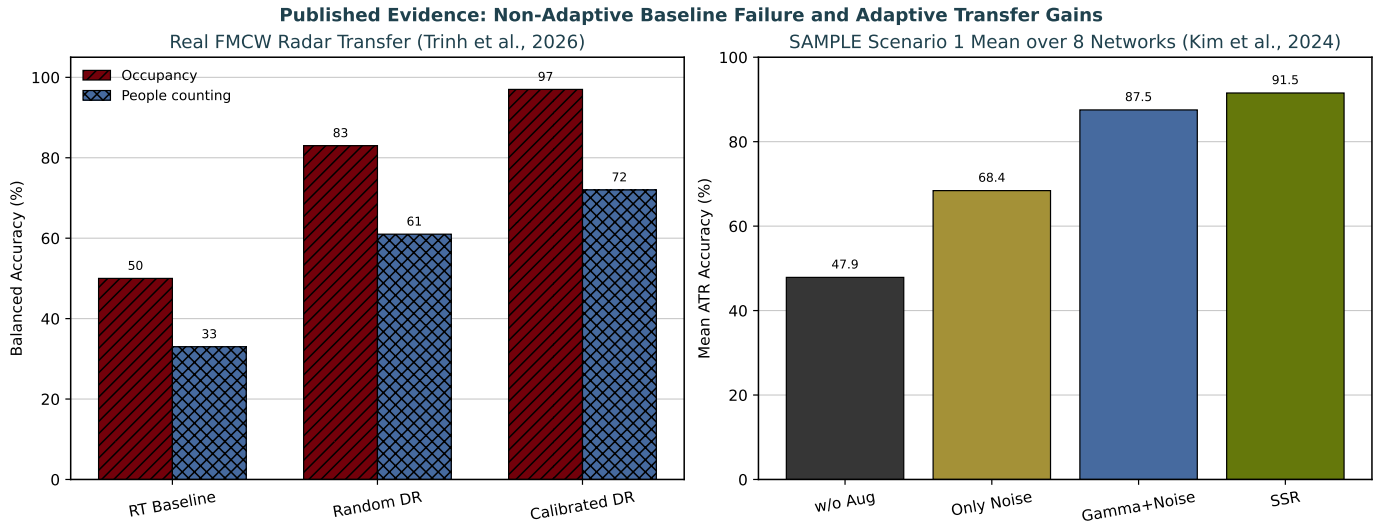


Fig. 3. Published quantitative evidence motivating Proposal 1. Left: non-adaptive and weakly adaptive baselines can fail under sim-to-real radar transfer; calibrated/adaptive methods improve substantially. Right: synthetic-only and simple noise baselines underperform on measured-data transfer compared to stronger adaptation strategies.

RL tuning is unstable, a contextual-bandit or Bayesian fallback still yields a complete and informative outcome.

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